ALY 6010 Probability theory and Intro Statistics

# Module 3 R practice

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## Introduction

This dataset comprises measurements and attributes related to 13,611 different dry beans, categorizing them into seven types: Seker, Barbunya, Bombay, Cali, Dermosan, Horoz, and Sira. These characteristics include details such as the area, perimeter, major and minor axis length, eccentricity, convex area, equivalent diameter, extent, solidity, roundness, compactness, and various shape factors of the beans. The report will involve conducting hypothesis testing, specifically one-sample t-tests and null hypothesis testing, on three variables: perimeter, major axis length, and minor axis length of the dry beans. These tests aim to evaluate whether the mean values of these variables differ significantly from certain expected values, providing insights into the characteristics of the beans. Additionally, hypothesis testing for p-values will be conducted for each of these variables to further analyze their significance and inferential implications.

**Reading Dataset** : Importing the file on dataset in R and reading dataset

```
library(readxl)
file_path <- "Dry_Bean_Dataset.xlsx"
Dry_Bean_Data <- read_excel(file_path)
head(Dry_Bean_Data)
```

```
## # A tibble: 6 × 17
##    Area Perimeter MajorAxisLength MinorAxisLength AspectRation Eccentricit
y
##   <dbl>     <dbl>           <dbl>           <dbl>        <dbl>        <dbl
>
## 1 28395      610.            208.            174.         1.20         0.55
0
## 2 28734      638.            201.            183.         1.10         0.41
2
## 3 29380      624.            213.            176.         1.21         0.56
3
## 4 30008      646.            211.            183.         1.15         0.49
9
## 5 30140      620.            202.            190.         1.06         0.33
4
## 6 30279      635.            213.            182.         1.17         0.52
0
## # ℹ 11 more variables: ConvexArea <dbl>, EquivDiameter <dbl>, Extent <dbl
>,
## #   Solidity <dbl>, roundness <dbl>, Compactness <dbl>, ShapeFactor1 <dbl>
,
## #   ShapeFactor2 <dbl>, ShapeFactor3 <dbl>, ShapeFactor4 <dbl>, Class <chr
>
```

## Installing Libraries

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(psych)

##
## Attaching package: 'psych'

## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

## PART-1 (DATA ANALYSES)

### Cleaning data-

To ensure accuracy, consistency, and reliability for analysis purposes. It eliminates errors, inconsistencies, duplicates, and missing values, and  improving data quality and facilitating more efficient analysis processes.

**Transforming decimal points into their minimal decimal increments**.

```
Dry_Bean_Data$Eccentricity <- round(Dry_Bean_Data$Eccentricity, 1)
Dry_Bean_Data$AspectRation <- round(Dry_Bean_Data$AspectRation, 1)
Dry_Bean_Data$Solidity <- round(Dry_Bean_Data$Solidity, 2)
Dry_Bean_Data$roundness <- round(Dry_Bean_Data$roundness, 2)
Dry_Bean_Data$Extent <- round(Dry_Bean_Data$Extent, 2)
Dry_Bean_Data$Compactness <- round(Dry_Bean_Data$Compactness, 2)
Dry_Bean_Data$Solidity <- round(Dry_Bean_Data$Solidity, 2)
Dry_Bean_Data$ShapeFactor1 <- round(Dry_Bean_Data$ShapeFactor1, 4)
Dry_Bean_Data$ShapeFactor2 <- round(Dry_Bean_Data$ShapeFactor2, 4)
Dry_Bean_Data$ShapeFactor3 <- round(Dry_Bean_Data$ShapeFactor3, 3)
Dry_Bean_Data$ShapeFactor4 <- round(Dry_Bean_Data$ShapeFactor4, 3)
```

### Converting value into integers

```
Dry_Bean_Data$Perimeter <- as.integer(Dry_Bean_Data$Perimeter)
Dry_Bean_Data$MajorAxisLength <- as.integer(Dry_Bean_Data$MajorAxisLength)
Dry_Bean_Data$MinorAxisLength <- as.integer(Dry_Bean_Data$MinorAxisLength)
Dry_Bean_Data$EquivDiameter <- as.integer(Dry_Bean_Data$EquivDiameter)
head(Dry_Bean_Data)

## # A tibble: 6 × 17
##    Area Perimeter MajorAxisLength MinorAxisLength AspectRation Eccentricit
y
##   <dbl>     <int>           <int>           <int>        <dbl>        <dbl
>
## 1 28395       610             208             173          1.2           0.
5
## 2 28734       638             200             182          1.1           0.
4
## 3 29380       624             212             175          1.2           0.
6
## 4 30008       645             210             182          1.2           0.
5
## 5 30140       620             201             190          1.1           0.
3
## 6 30279       634             212             181          1.2           0.
5
## # ℹ 11 more variables: ConvexArea <dbl>, EquivDiameter <int>, Extent <dbl
>,
## #   Solidity <dbl>, roundness <dbl>, Compactness <dbl>, ShapeFactor1 <dbl>
```

```
,
## #   ShapeFactor2 <dbl>, ShapeFactor3 <dbl>, ShapeFactor4 <dbl>, Class <chr
>
```

### Identifying the class of the columns

```
Dry_Bean_Data_class <- sapply(Dry_Bean_Data, class)
Dry_Bean_Data_class

##            Area       Perimeter MajorAxisLength MinorAxisLength    AspectR
ation
##       "numeric"       "numeric"       "numeric"       "numeric"       "num
eric"
##    Eccentricity      ConvexArea   EquivDiameter          Extent         Sol
idity
##       "numeric"       "numeric"       "numeric"       "numeric"       "num
eric"
##       roundness     Compactness    ShapeFactor1    ShapeFactor2    ShapeFa
ctor3
##       "numeric"       "numeric"       "numeric"       "numeric"       "num
eric"
##    ShapeFactor4           Class
##       "numeric"     "character"
```

### Grouping 7 different beans

Grouping seven different beans for organizing and categorizing the data according to their shared common characteristics.

```
class_count <- Dry_Bean_Data %>% group_by(Class) %>%
  summarise(class_count = n())
class_count

## # A tibble: 7 × 2
##   Class    class_count
##   <chr>          <int>
## 1 BARBUNYA        1322
## 2 BOMBAY           522
## 3 CALI            1630
## 4 DERMASON        3546
## 5 HOROZ           1928
## 6 SEKER           2027
## 7 SIRA            2636
```

## Summary of dataset

The dataset summary offers a comprehensive overview, presenting key statistical measures such as the mean, mode, median, standard deviation, minimum, and maximum values. This summary provides valuable insights into the central tendency, dispersion, and range of the data.

```
summary(Dry_Bean_Data)

##       Area          Perimeter      MajorAxisLength MinorAxisLength
##  Min.   : 20420   Min.   : 524.0   Min.   :183.0   Min.   :122.0  
##  1st Qu.: 36328   1st Qu.: 703.0   1st Qu.:253.0   1st Qu.:175.0  
##  Median : 44652   Median : 794.0   Median :296.0   Median :192.0  
##  Mean   : 53048   Mean   : 854.8   Mean   :319.6   Mean   :201.8  
##  3rd Qu.: 61332   3rd Qu.: 977.0   3rd Qu.:376.0   3rd Qu.:217.0  
##  Max.   :254616   Max.   :1985.0   Max.   :738.0   Max.   :460.0  
##   AspectRation    Eccentricity     ConvexArea     EquivDiameter  
##  Min.   :1.000   Min.   :0.200   Min.   : 20684   Min.   :161.0  
##  1st Qu.:1.400   1st Qu.:0.700   1st Qu.: 36714   1st Qu.:215.0  
##  Median :1.600   Median :0.800   Median : 45178   Median :238.0  
##  Mean   :1.583   Mean   :0.753   Mean   : 53768   Mean   :252.6  
##  3rd Qu.:1.700   3rd Qu.:0.800   3rd Qu.: 62294   3rd Qu.:279.0  
##  Max.   :2.400   Max.   :0.900   Max.   :263261   Max.   :569.0  
##      Extent          Solidity        roundness       Compactness    
##  Min.   :0.5600   Min.   :0.9200   Min.   :0.4900   Min.   :0.6400  
##  1st Qu.:0.7200   1st Qu.:0.9900   1st Qu.:0.8300   1st Qu.:0.7600  
##  Median :0.7600   Median :0.9900   Median :0.8800   Median :0.8000  
##  Mean   :0.7498   Mean   :0.9875   Mean   :0.8733   Mean   :0.7999  
##  3rd Qu.:0.7900   3rd Qu.:0.9900   3rd Qu.:0.9200   3rd Qu.:0.8300  
##  Max.   :0.8700   Max.   :0.9900   Max.   :0.9900   Max.   :0.9900  
##   ShapeFactor1       ShapeFactor2       ShapeFactor3     ShapeFactor4   
##  Min.   :0.002800   Min.   :0.000600   Min.   :0.4100   Min.   :0.9480  
##  1st Qu.:0.005900   1st Qu.:0.001200   1st Qu.:0.5810   1st Qu.:0.9940  
##  Median :0.006600   Median :0.001700   Median :0.6420   Median :0.9960  
##  Mean   :0.006564   Mean   :0.001716   Mean   :0.6436   Mean   :0.9951  
##  3rd Qu.:0.007300   3rd Qu.:0.002200   3rd Qu.:0.6960   3rd Qu.:0.9980  
##  Max.   :0.010500   Max.   :0.003700   Max.   :0.9750   Max.   :1.0000  
##     Class          
##  Length:13611      
##  Class :character  
##  Mode  :character  
##                    
##                    
## 
```

# Constructing one sample t-test for dry bean data

### a) No. of observations

```
no.of.observations <- nrow(Dry_Bean_Data)
cat("Number of observations in the Dry Bean Data: ", no.of.observations)

## Number of observations in the Dry Bean Data:  13611
```

### b) Degree of freedom

```
degrees.of.freedom <- no.of.observations -1
cat("Degree of freedom :",degrees.of.freedom)

## Degree of freedom : 13610
```

### c) significance level

```
significance_level <- 0.05
```

### d) lower critical value

```
lower.critical.value <-
  qt( significance_level / 2, degrees.of.freedom )
cat( "Lower critical value:",
     round( lower.critical.value, 5 ) )

## Lower critical value: -1.96014
```

### e) upper critical value

```
upper.critical.value <-
  qt( 1 - significance_level / 2, degrees.of.freedom )
cat( "Upper critical value:",
     round( upper.critical.value, 5 ) )

## Upper critical value: 1.96014
```

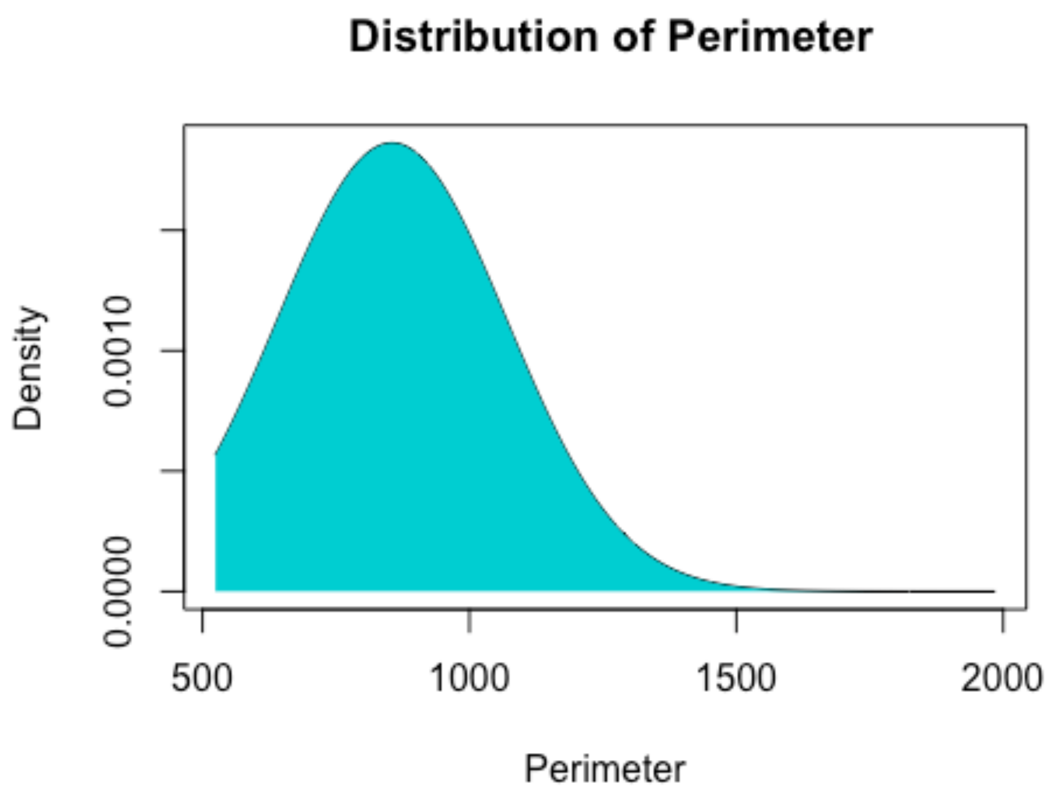
# Conducting One sample t-test

## 1) Selected variable "Perimeter" for further analysis

```
Perimeter <- Dry_Bean_Data$Perimeter
```

### Distribution of Perimeter

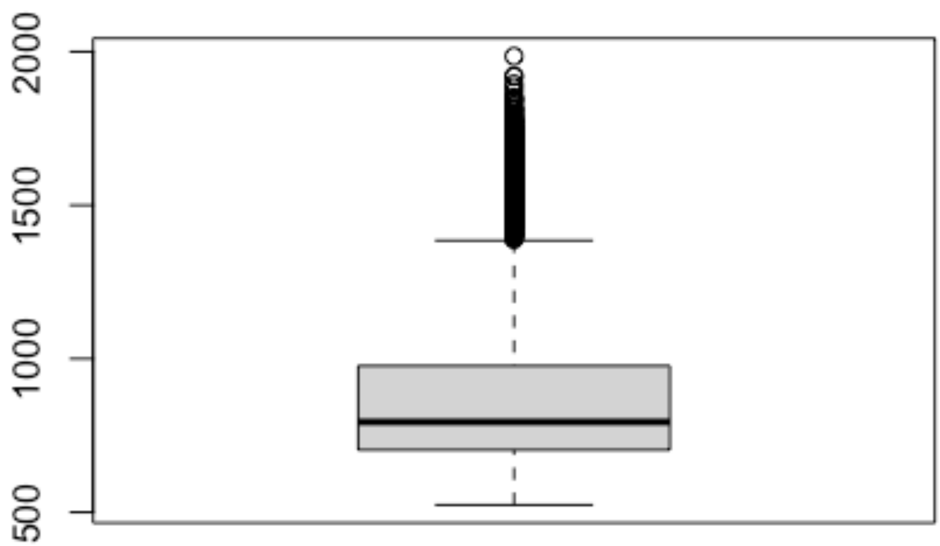
```
mean_perimeter <- mean(Dry_Bean_Data$Perimeter)
sd_perimeter <- sd(Dry_Bean_Data$Perimeter)
x <- seq(min(Dry_Bean_Data$Perimeter), max(Dry_Bean_Data$Perimeter), length.o
ut = 1000)
y <- dnorm(x, mean = mean_perimeter, sd = sd_perimeter)
plot(x, y, type = "l", main = " Distribution of Perimeter", xlab = "Perimeter
", ylab = "Density")
polygon(c(min(x), x, max(x)), c(0, y, 0), col = "darkturquoise", border = NA)
```




**Description:** The distribution of perimeter data exhibits a right-skewed pattern, indicating that most values are concentrated towards the lower end of the range, with a few larger values extending the tail towards the higher end. This skewness suggests that there are relatively fewer instances of larger perimeter values compared to smaller ones.

**Perimeter box plot**

```
boxplot(Dry_Bean_Data$Perimeter)
```




**Calculate the variance of the Perimeter variable**

Calculating the variance of the Perimeter variable for understanding the spread or dispersion of data points around the mean. Variance quantifies the average squared deviation of each data point from the mean, offering a measure of how much values in the dataset differ from the average value. This computation provides insights into the variability or scatter of the perimeter measurements, crucial for statistical analysis and making inferences about the dataset.

```
variance_perimeter <- var(Perimeter)
cat("The variance of the Perimeter variable is:", round(variance_perimeter, 5
))

## The variance of the Perimeter variable is: 45922.27
```

## one sample t -test

**Null Hypothesis (H0):** The null hypothesis states that the mean perimeter of dry beans is equal to 800, $\mu 0 = 800$.

H0: $\mu = \mu 0$

**Alternative Hypothesis (Ha):** The alternative hypothesis suggests that the mean perimeter of dry beans is not equal to 800.

Ha: $\mu \neq \mu 0$ (two-tailed test)

Collecting the "Perimeter" data from the dry bean dataset, a one-sample t-test is conducted using the collected "Perimeter" data and the specified null hypothesis expected value ($\mu 0$ = 800).

```
#H0:/mu=800
null_hypothesis_expected_value <- 800
t_test_result <- t.test (Perimeter, mu = null_hypothesis_expected_value)
print(t_test_result)
```

```
##
##  One Sample t-test
##
## data:  Perimeter
## t = 29.825, df = 13610, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 800
## 95 percent confidence interval:
##  851.1819 858.3827
## sample estimates:
## mean of x
##  854.7823
```

## P-value of Perimeter

The p-value associated with the perimeter data represents the probability of obtaining the observed results, if the null hypothesis were true. P-value <2.2 which is less than significance level

### Result of one sample t-test

The test resulted in a test statistic (t-value) of 29.825 with a degree of freedom (df) of 13610. The calculated **p-value** was found to be less than 2.2, which is significantly smaller than the chosen significance level of 0.5.

The null hypothesis (H0) says the mean perimeter of dry beans is 800. The alternative hypothesis (H1) says it's not 800.

Since the p-value is so small, we reject the null hypothesis. This means there's enough evidence to say the mean perimeter of dry beans is significantly different from 800.

Also, the 95 percent confidence interval for the mean perimeter is between 851.1819 and 858.3827, with a sample estimate mean of 854.7823. This interval tells us where we can be 95 percent confident the true mean perimeter of dry beans lies.

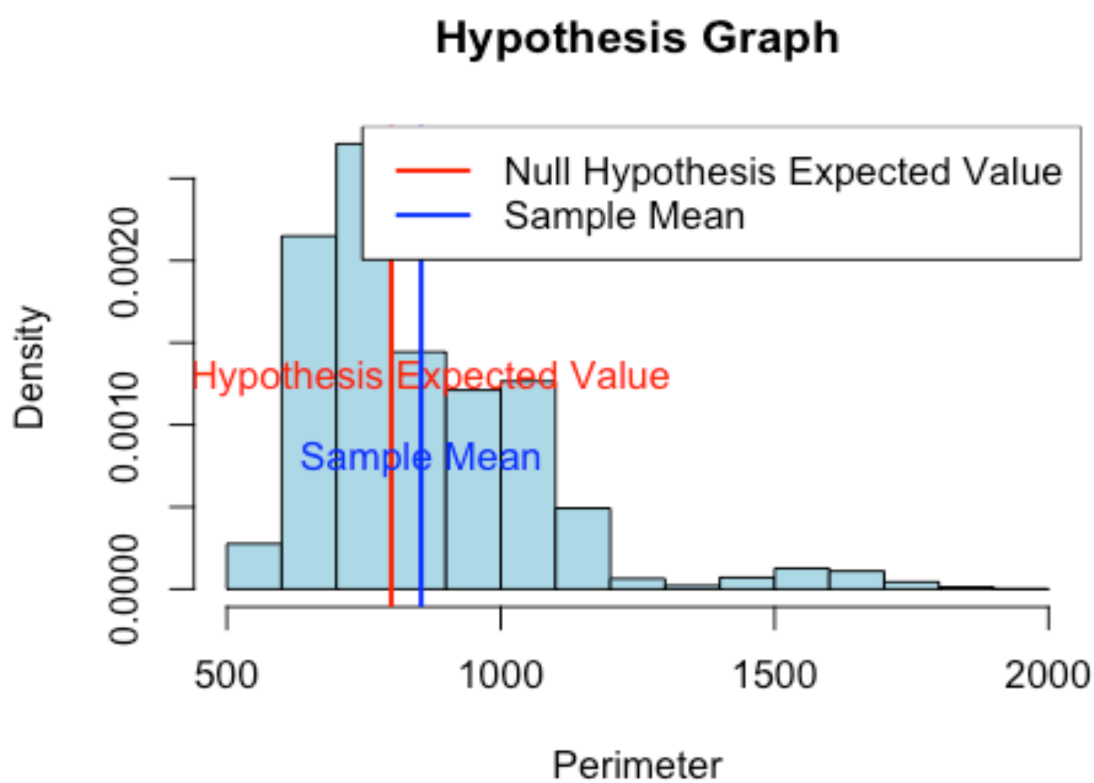
### Perimeter mean

```
mean_perimeter <- mean(Dry_Bean_Data$Perimeter)
print(mean_perimeter)

## [1] 854.7823
```

**Graph of hypothesis testing**

```
null_hypothesis_expected_value <- 800
sample_mean <- t_test_result$estimate
hist(Perimeter, freq = FALSE, main = "Hypothesis Graph", xlab = "Perimeter",
ylab = "Density", col = "lightblue")
abline(v = null_hypothesis_expected_value, col = "red", lwd = 2)
text(null_hypothesis_expected_value, 0.0015, "Null Hypothesis Expected Value"
, pos = 1, col = "red")
abline(v = sample_mean, col = "blue", lwd = 2)
text(sample_mean, 0.001, "Sample Mean", pos = 1, col = "blue")
legend("topright", legend = c("Null Hypothesis Expected Value", "Sample Mean"
), col = c("red", "blue"), lwd = 2)
```



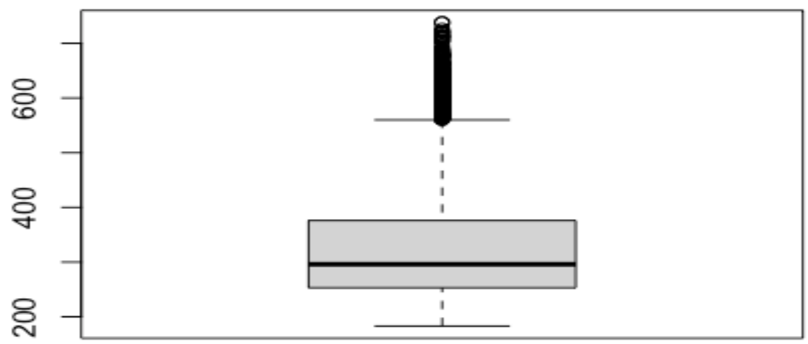

**Description:** The graph visually represents the hypothesis testing conducted for the perimeter of dry beans. It likely illustrates the distribution of perimeter measurements along with the mean perimeter and the 95 percent confidence interval. Given the statistical results obtained from the one-sample t-test, which include a high test statistic (t-value), an extremely low p-value, and a confidence interval that does not include the null hypothesis value of 800, the graph likely reflects these findings.

## 2) **Selected variable "Major Axis Length" for further analysis**

```
major_axis_length <- Dry_Bean_Data$MajorAxisLength
```

**Plotting Major Axis lenght**

```
boxplot(Dry_Bean_Data$MajorAxisLength)
```




**Conducting one sample t-test in major axis length**

**Null Hypothesis (H0):** The null hypothesis states that the mean major axis length of dry beans is equal to 300, μ0 = 300.

H0: μ = μ0

**Alternative Hypothesis (Ha):** The alternative hypothesis suggests that the mean major axis length of dry beans is not equal to 300.

Ha: μ ≠ μ0 (two-tailed test)

Collecting the "Major Axis Length" data from the dry bean dataset, a one-sample t-test is conducted using the collected "Perimeter" data and the specified null hypothesis expected value (μ0 = 300).

```
null_hypothesis_expected_value <- 300
t_test_result <- t.test(major_axis_length, mu = null_hypothesis_expected_valu
e)
print(t_test_result)

##
##  One Sample t-test
```

```
##
## data:  major_axis_length
## t = 26.742, df = 13610, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 300
## 95 percent confidence interval:
##  318.2024 321.0820
## sample estimates:
## mean of x
##  319.6422
```

### P-value of major axis length

The p-value associated with the major axis length data represents the probability of obtaining the observed results, if the null hypothesis were true. P-value <2.2 which is less than significance level

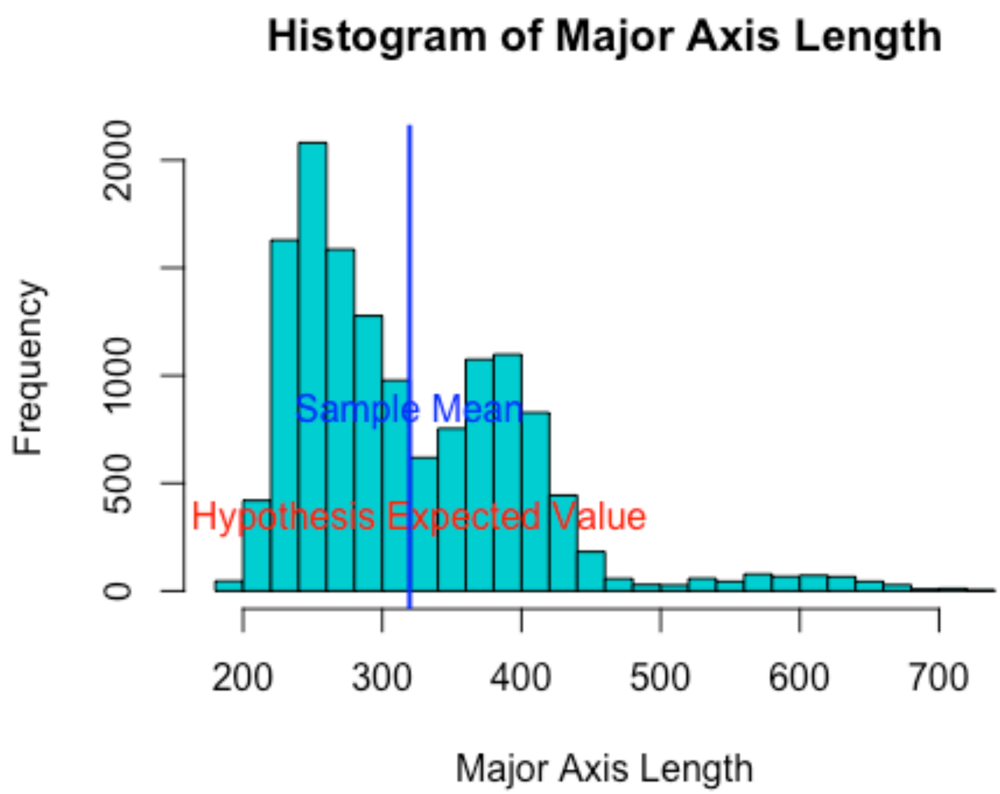
### Result of one sample t-test

The p-value obtained from the one-sample t-test for the major axis length data is extremely low (< 2.2), indicating strong evidence against the null hypothesis. This suggests that the true mean major axis length is significantly different from the expected value of 300. The alternative hypothesis, which proposes that the true mean is not equal to 300, is supported by the data.

Moreover, the 95 percent confidence interval for the mean major axis length, ranging from 318.2024 to 321.0820, provides a range within which we can be 95 percent confident that the true mean major axis length lies. This interval does not include the null hypothesis value of 300, further supporting the rejection of the null hypothesis.

**Graph Major axis length**

```
null_hypothesis_expected_value <- 300
sample_mean <- t_test_result$estimate
hist(major_axis_length, breaks = 30, main = "Histogram of Major Axis Length",
xlab = "Major Axis Length", ylab = "Frequency", col = "darkturquoise")
text(null_hypothesis_expected_value, 500, "Null Hypothesis Expected Value", p
os = 1, col = "red")
abline(v = sample_mean, col = "blue", lwd = 2)
text(sample_mean, 1000, "Sample Mean", pos = 1, col = "blue")
```



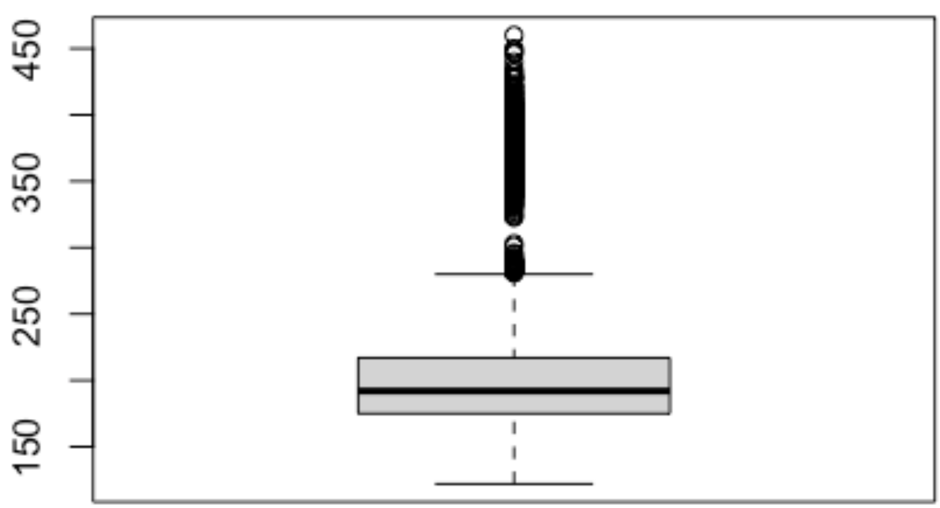

**Description**: The graph visually represents the hypothesis testing conducted for the Major axis length of dry beans. It likely illustrates the distribution of perimeter measurements along with the mean major axis length and the 95 percent confidence interval. Given the statistical results obtained from the one-sample t-test, which include a high test statistic (t-value), an extremely low p-value, and a confidence interval that does not include the null hypothesis value of 300, the graph likely reflects these findings.

### 3) Selected variable Minor axis length

```
minor_axis_length <- Dry_Bean_Data$MinorAxisLength
```

**Plotting Major Axis lenght**

```
boxplot(Dry_Bean_Data$MinorAxisLength)
```




## one sample t -test

**Null Hypothesis (H0):** The null hypothesis states that the mean minor axis length of dry beans is equal to 250, μ0 = 250.

H0: μ = μ0

**Alternative Hypothesis (Ha):** The alternative hypothesis suggests that the mean minor axis length of dry beans is not equal to 250.

Ha: μ ≠ μ0 (two-tailed test)

Collecting the " Minor axis length " data from the dry bean dataset, a one-sample t-test is conducted using the collected " Minor axis length " data and the specified null hypothesis expected value (μ0 = 250).

```
null_hypothesis_expected_value <- 250
t_test_result <- t.test(minor_axis_length, mu = null_hypothesis_expected_valu
e)
print(t_test_result)
```

```
##
##  One Sample t-test
##
## data:  minor_axis_length
## t = -125.11, df = 13610, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 250
## 95 percent confidence interval:
##  201.0181 202.5292
## sample estimates:
## mean of x
##  201.7736
```

### P-value of minor axis length

The p-value associated with the minor axis length data represents the probability of obtaining the observed results, if the null hypothesis were true. P-value <2.2 which is less than significance level.

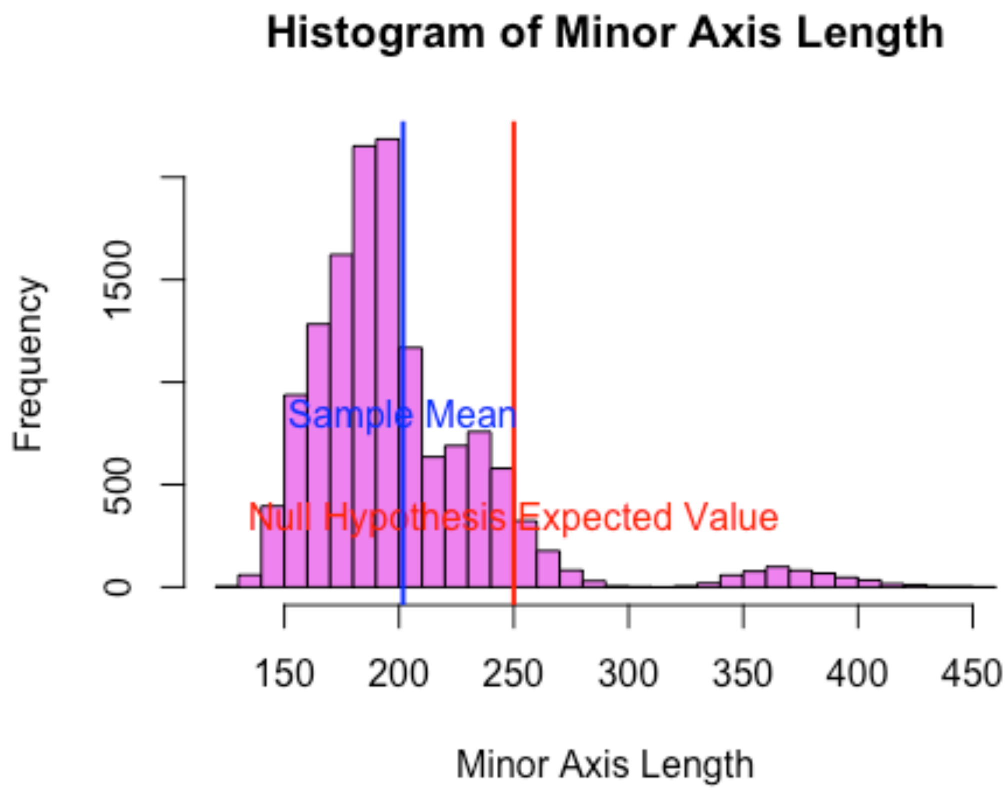
### Result of one sample t-test

The p-value obtained from the one-sample t-test for the minor axis length data is extremely low (< 2.2), indicating strong evidence against the null hypothesis. This suggests that the true mean minor axis length is significantly different from the expected value of 250. The alternative hypothesis, which proposes that the true mean is not equal to 250, is supported by the data.

Additionally, the 95 percent confidence interval for the mean minor axis length ranges from 201.0181 to 202.5292, providing a range within which we can be 95 percent confident that the true mean minor axis length lies. Overall, these results indicate a significant difference in the mean minor axis length compared to the expected value of 250.

**Hypothesis testing plot for Minor axis length**

```
null_hypothesis_expected_value <- 250
sample_mean <- t_test_result$estimate
hist(minor_axis_length, breaks = 30, main = "Histogram of Minor Axis Length",
xlab = "Minor Axis Length", ylab = "Frequency", col = "violet")
abline(v = null_hypothesis_expected_value, col = "red", lwd = 2)
text(null_hypothesis_expected_value, 500, "Null Hypothesis Expected Value", p
os = 1, col = "red")
abline(v = sample_mean, col = "blue", lwd = 2)
text(sample_mean, 1000, "Sample Mean", pos = 1, col = "blue")
```




**Description**: The graph visually represents the hypothesis testing conducted for the Major axis length of dry beans. It likely illustrates the distribution of perimeter measurements along with the mean major axis length and the 95 percent confidence interval. Given the statistical results obtained from the one-sample t-test, which include a high test statistic (t-value), an extremely low p-value, and a confidence interval that does not include the null hypothesis value of 250, the graph likely reflects these findings.

**Summary:**

In the analysis of the dry bean dataset, significant insights were uncovered regarding the perimeter, major axis length, and minor axis length of the beans.

For the Perimeter, the mean measurement was approximately 854.78 units, indicating a considerable spread of values. A one-sample t-test revealed a test statistic (t-value) of 29.825 with an extremely low p-value (< 2.2), suggesting strong evidence against the null hypothesis and signifying a significant difference in the mean perimeter compared to the expected value.

Similarly, for the Major Axis Length, the mean measurement was approximately 319.64 units, showing variability across the dataset. A one-sample t-test yielded a test statistic (t-value) of 26.742 with an extremely low p-value (< 2.2), indicating a significant difference in the mean major axis length compared to the expected value.

Regarding the Minor Axis Length, the mean measurement was approximately 201.77 units, demonstrating variability akin to the other attributes. A one-sample t-test produced a test statistic (t-value) of -125.11 with an extremely low p-value (< 2.2), signifying a significant difference in the mean minor axis length compared to the expected value.